- In a random experiment, the set of all outcomes are known but exact outcome is not known.
- The set of all possible outcomes is called sample space.
- A, B are said to be mutually exclusive events if $A \cap B = \phi$
- Probability of event E is $P(E) = \dfrac{n(E)}{n(S)}$

  (i) The probability of sure event is 1 and the probability of impossible event is 0.
  (ii) $0 \leq P(E) \leq 1$; (iii) $P(\bar{E}) = 1 - P(E)$
- If A and B are mutually exclusive events then $P(A \cup B) = P(A) + P(B)$.
- (i) $P(A \cap \bar{B}) = P(\text{only } A) = P(A) - P(A \cap B)$

  (ii) $P(\bar{A} \cap B) = P(\text{only } B) = P(B) - P(A \cap B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, for any two events A, B.
- For any three events A, B, C

  $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C)$
  $-P(C \cap A) + P(A \cap B \cap C)$

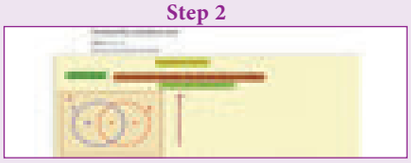
ICT CORNER

ICT 8.1

Step 1: Open the Browser type the URL Link given below (or) Scan the QR Code. Chapter named **"Probability"** will open. Select the work sheet **" Probability Addition law "**

Step 2: In the given worksheet you can change the question by clicking on "New Problem". Move the slider to see the steps.

Step 1 | Step 2 | Expected results

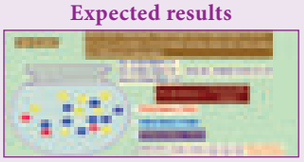

ICT 8.2

Step 1: Open the Browser type the URL Link given below (or) Scan the QR Code. Chapter named **"Probability"** will open. Select the work sheet **" Addition law Mutually Exclusive"**

Step 2: In the given worksheet you can change the question by clicking on "New Problem". Click on the check boxes to see the respective answer.

Step 1 | Step 2 | Expected results





You can repeat the same steps for other activities

https://www.geogebra.org/m/jfr2zzgy#chapter/359554
or Scan the QR Code.

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